Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. Which of the following must be true for a linear Kalman filter to be stable? Select all that apply.

1 / 1 point

- [x] For a linear Kalman filter to be stable, $\Sigma^+_{\tilde{x},k}$ must always be positive semi-definite (it must have non-negative eigenvalues).

  Correct
  Yes, this is true.

- [ ] For a linear Kalman filter to be stable, the system whose state is being estimated must be unstable.
- [x] For a linear Kalman filter to be stable, $\Sigma^+_{\tilde{x},k}$ must always be a symmetic matrix.

  Correct
  Yes, this is true.

- [ ] For a linear Kalman filter to be stable, both the process noise w_k and the sensor noise v_k must be small.
- [ ] For a linear Kalman filter to be stable, its computation of $\hat{x}^+_k$ must always be equal to the true x_k.

2. What is the primary motivation behind the development of the square-root Kalman filter?

1 / 1 point

- ○ It is faster than the standard Kalman filter (i.e., it requires fewer floating-point operations per iteration).
- ◉ It reduces the number of bits per variable required to estimate the state to a desired precision.
- ○ It forces all covariance matrices to be positive definite.
- ○ It provides more accurate estimates than the standard Kalman filter.

  Correct
  Yes, this is true. Alternately, it increases the precision for a given number of bits of storage per variable.

3. Run the Kalman-filter and square-root Kalman-filter code in the Coursera Labs Jupyter notebook for Lesson 2.2.2. What do you observe?

1 / 1 point

- ◉ There is no noticeable difference in results between the two filters.
- ○ The standard Kalman filter is noticeably less accurate than the square-root Kalman filter.
- ○ The square-root Kalman filter is noticeably less accurate than the standard Kalman filter.
- ○ There are noticeable differences near the beginning of the simulation, but after some time the square-root Kalman filter converges to the standard Kalman-filter solution.

  Correct
  Yes. When using double-precision floating-point variables in Octave, there is no noticeable difference in results.

4. If we do not believe that the true initial state of the system x_0 is inside the error bounds $\hat{x}^+_0 \pm 3\sqrt{\mathrm{diag}(\Sigma^+_{\tilde{x},0})}$, what should we do?

1 / 1 point

- ○ We should increase $\Sigma_{\tilde{v}}$.
- ○ We should decrease $\Sigma_{\tilde{x},0}$.
- ○ We should decrease $\Sigma_{\tilde{w}}$.
- ◉ We should increase $\Sigma_{\tilde{x},0}$.
- ○ We should increase $\Sigma_{\tilde{w}}$.
- ○ We should decrease $\Sigma_{\tilde{v}}$.

  Correct
  Yes. This is the correct approach to tuning the initial state error covariance.

5. If we believe that our linear Kalman filter state estimate is converging more slowly than it could, what should we try?

1 / 1 point

- ○ We should increase $\Sigma_{\tilde{w}}$.
- ◉ We should decrease $\Sigma_{\tilde{w}}$.
- ○ We should increase $\Sigma^+_{\tilde{x},0}$.
- ○ We should decrease $\Sigma^+_{\tilde{x},0}$.
- ○ We should increase $\Sigma_{\tilde{v}}$.

  Correct
  Yes. This will cause the filter to believe that there is less process noise, leading to reduced state uncertainty in step 1b, and hence faster convergence.

6. What strategy is proposed this week to estimate a system's state vector when the process and/or sensor noises have a dc bias?

1 / 1 point

- ○ Execute a standard Kalman filter without making any changes. It works well even when the noises have dc bias.
- ○ Execute all the steps of a standard Kalman filter plus some additional steps to estimate the biases. Then, subtract the biases from w_k and v_k in the standard Kalman filter steps.
- ◉ Execute all the steps of a standard Kalman filter plus some additional steps to estimate the biases. Then, adjust the Kalman-filter output to compensate for the biases.
- ○ Implement a high-pass filter to "notch out" the dc bias from the measurements and add these filter equations to the standard Kalman filter.

  Correct
  Yes, this is correct.

7. Execute the Coursera Labs Jupyter notebook for Lesson 2.2.4. What is the percentage of time that the **velocity**-state error is outside of the estimation-error bounds when the filter is **IS** corrected for bias? Enter your answer with one digit to the right of the decimal point.

1 / 1 point

1.7

Correct
Yes. That is correct.

8. When process and sensor noises are correlated **at the same time instant**, what strategy do we take to make unbiased state estimates? (Select all that apply.)

1 / 1 point

- [x] We modify Kalman-filter Step 1a with a new $\bar{A}_k$ matrix and an augmented deterministic input.

  Correct
  Yes. This is correct.

- [x] We modify Kalman-filter Step 1b to use a new $\Sigma_{\tilde{w}}$ matrix instead of $\Sigma_{\tilde{w}}$.

  Correct
  Yes. This is correct.

- [ ] We modify how we compute L_k in Kalman-filter Step 2a.
- [ ] We implement a different computation for $\hat{x}^+_k$ in Kalman-filter Step 2b.
- [ ] We compute a different value for $\Sigma^+_{\tilde{x},k}$ in Kalman-filter Step 2c because of the change in how we compute L_k.

9. When process and sensor noises are correlated **at a time offset of one sample**, what strategy do we take to make unbiased state estimates? (Select all that apply.)

1 / 1 point

- [ ] We modify Kalman-filter Step 1a with a new $\bar{A}_k$ matrix and an augmented deterministic input.
- [ ] We modify Kalman-filter Step 1b to use a new $\Sigma_{\tilde{w}}$ matrix instead of $\Sigma_{\tilde{w}}$.
- [x] We modify how we compute L_k in Kalman-filter Step 2a.

  Correct
  Yes. This is correct.

- [x] We compute a different value for $\hat{x}^+_k$ in Kalman-filter Step 2b because of the change in how we compute L_k.

  Correct
  Yes. This is correct.

- [x] We compute a different value for $\Sigma^+_{\tilde{x},k}$ in Kalman-filter Step 2c because of the change in how we compute L_k.

  Correct
  Yes. This is correct.

10. Execute the Coursera Labs Jupyter notebook for Lesson 2.2.6. Consider the case where the **process** noise is autocorrelated (not white) and a **standard** Kalman filter is implemented (with no compensation for the fact that process noise is autocorrelated).

For this case, what is the percentage of time that the **velocity** estimate is outside of the filter's confidence bounds? Enter your result with one digit to the right of the decimal point.

1 / 1 point

40.1

Correct
Yes. This is correct.